

National Health and Nutrition Examination Survey

August 2021-August 2023 Data Documentation, Codebook, and Frequencies

Perfluoroalkyl and Polyfluoroalkyl Substances (PFAS_L)

Data File: PFAS_L.xpt

First Published: August 2026

Last Revised: NA

Component Description

Perfluoroalkyl and polyfluoroalkyl substances (PFAS) are used in multiple commercial applications, including surfactants, lubricants, paints, polishes, food packaging, and fire-retarding foams. Certain PFAS are used in the manufacture of polymers used in many industrial and consumer products, including soil, stain, grease, and water-resistant coatings on textiles and carpet; uses in the automotive, mechanical, aerospace, chemical, electrical, medical, and building/construction industries; personal care products; and non-stick coatings on cookware. Some PFAS are ubiquitous contaminants found in humans and animals worldwide.

Synthesis of PFAS employed electrochemical fluorination (ECF) or fluorotelomerization. ECF, used from the 1950s until the early 2000s, yielded branched and linear isomers. By contrast, fluorotelomerization produces almost exclusively linear compounds (Vyas, et. al., 2007). The structural isomer patterns of perfluorooctanoate (PFOA) and perfluorooctane sulfonate (PFOS) in humans may be useful for understanding routes and sources of exposure (Benskin, et. al., 2010). Therefore, concentrations of linear PFOA (n-PFOA), the sum of branched isomers of PFOA (Sb-PFOA, branched PFOA isomers), linear PFOS (n-PFOS), and the sum of perfluoromethylheptane sulfonate isomers (Sm-PFOS, monomethyl branched PFOS isomers) were measured in serum.

Eligible Sample

Examined participants aged 12 years and older were eligible.

Description of Laboratory Methodology

Online solid phase extraction coupled to high performance liquid chromatography-turboionspray ionization-tandem mass spectrometry (online SPE-HPLC-TIS-MS/MS) is used for the quantitative detection of PFAS: 2-(N-methyl-perfluorooctane sulfonamido) acetate (Me-PFOA-AcOH), perfluorohexane sulfonate (PFHxS), n-perfluorooctane sulfonate (n-PFOS), the sum of perfluoromethylheptane sulfonate isomers (Sm-PFOS, monomethyl branched isomers of PFOS), n-perfluorooctanoate (n-PFOA), the sum of branched perfluorooctanoate isomers (Sb-PFOA, branched PFOA isomers), perfluorononanoate (PFNA), perfluorodecanoate (PFDeA), perfluoroundecanoate (PFUA), and perfluorododecanoate (PFDoA) 21. Briefly, after dilution with formic acid, one aliquot of 50 µL of serum is injected into a commercial column switching system allowing for concentration of the analytes on solid-phase extraction column. Separation of the analytes from each other and from other serum components is achieved with high-performance liquid chromatography. Detection and quantification are done using negative-ion TurboIonSpray ionization, a variant of electrospray ionization, tandem mass spectrometry. This method allows for rapid detection of these PFAS in human serum with limits of detection in the low parts per billion (ppb or ng/mL) range.

Refer to the Laboratory Method Files section for a detailed description of the laboratory methods used.

Laboratory Method Files

[Perfluoroalkyl and Polyfluoroalkyl Substances](#) (August 2026)

Laboratory Quality Assurance and Monitoring

Serum specimens were processed, stored, and shipped to the Division of Laboratory Sciences, National Center for Environmental Health, Centers for Disease Control and Prevention, Atlanta, GA for analysis.

Detailed instructions on specimen collection and processing are discussed in the [NHANES Laboratory Procedures Manual \(LPM\)](#). Vials were stored under appropriate frozen (-30°C) conditions until they are shipped to the National Center for Environmental Health for testing.

The NHANES quality assurance and quality control (QA/QC) protocols meet the 1988 Clinical Laboratory Improvement Act mandates. Detailed QA/QC instructions are discussed in the [NHANES LPM](#).

Mobile Examination Centers (MECs)

Laboratory team performance is monitored using several techniques. NCHS and contract consultants use a structured competency assessment evaluation during visits to evaluate both the quality of the laboratory work and the QC procedures. Each laboratory staff member is observed for equipment operation, specimen collection and preparation; testing procedures and constructive feedback are given to each staff member. Formal retraining sessions are conducted annually to ensure that required skill levels were maintained.

Analytical Laboratories

NHANES uses several methods to monitor the quality of the analyses performed by the contract laboratories. In the MEC, these methods include performing blind split samples collected during "dry run" sessions. In addition, contract laboratories randomly perform repeat testing on 2% of all specimens.

NCHS developed and distributed a QC protocol for all CDC and contract laboratories, which outlined the use of Westgard rules (Westgard, et al. 1981) when running NHANES specimens. Progress reports containing any problems encountered during shipping or receipt of specimens, summary statistics for each control pool, QC graphs, instrument calibration, reagents, and any special considerations are submitted to NCHS quarterly. The reports are reviewed for trends or shifts in the data. The laboratories are required to explain any identified areas of concern.

All QC procedures recommended by the manufacturers were followed. Reported results for all assays meet the Division of Laboratory Sciences' QA/QC performance criteria for accuracy and precision, similar to the Westgard rules (Caudill, et al., 2008).

Data Processing and Editing

The data were reviewed. Incomplete data or improbable values were sent to the performing laboratory for confirmation.

Analytic Notes

There are over 800 laboratory tests performed on NHANES participants. However, not all participants provided biospecimens or enough volume for all the tests to be performed. The specimen availability can also vary by age or other population characteristics. For example, in 2017-2018, approximately 80% of children aged 1-17 years who were examined in the MEC provided a blood specimen through phlebotomy, while 95% of examined adults age 18 and

older provided a blood specimen. Analysts should evaluate the extent of missing data in the dataset related to the outcome of interest as well as any predictor variables used in the analyses to determine whether additional re-weighting for item non-response is necessary.

Please refer to the NHANES [Analytic Guidelines](#) and the on-line NHANES [Tutorial](#) for details on the use of sample weights and other analytic issues.

Subsample weights

Serum PFAS were measured in a one-half subsample of participants 12 years and over. Special sample weights are required to analyze these data properly. Specific sample weights for this subsample are included in this data file and should be used when analyzing these data.

Demographic and Other Related Variables

The analysis of NHANES laboratory data must be conducted using the appropriate survey design and demographic variables. The NHANES [August 2021- August 2023 Demographic File](#) contains demographic data, health indicators, and other related information collected during household interviews as well as the sample design variables. The recommended procedure for variance estimation requires use of stratum and PSU variables (SDMVSTRA and SDMVPSU, respectively) in the demographic data file.

The [Fasting Questionnaire File](#) includes auxiliary information, such as fasting status, length of fast, and the time of venipuncture.

This laboratory data file can be linked to the other NHANES data files using the unique survey participant identifier (i.e., SEQN).

Detection Limits

The detection limits were constant for all of the analytes in the data set. Two variables are provided for each of these analytes. The variable name ending in "L" (ex., LBDPFHSL) indicates whether the result was below the limit of detection: the value "0" means that the result was at or above the limit of detection, "1" indicates that the result was below the limit of detection. For analytes with analytic results below the lower limit of detection (ex., LBDPFHSL=1), an imputed fill value was placed in the analyte results field. This value is the lower limit of detection divided by square root of 2 ($LLOD/\sqrt{2}$). The other variable prefixed LBX (ex., LBXPFS) provides the analytic result for that analyte.

The lower limit of detection (LLOD, in ng/mL) for each PFAS:

Variable Name	Analyte Description	LLOD
LBXPFDE	Perfluorodecanoic acid (PFDeA) (ng/mL)	0.10
LBXPFHS	Perfluorohexane sulfonic acid (PFHxS) (ng/mL)	0.10
LBXMPAH	2-(N-methylperfluorooctanesulfonamido)acetic acid (Me-PFOSA-AcOH) (ng/mL)	0.10
LBXPFNA	Perfluorononanoic acid (PFNA) (ng/mL)	0.10
LBXPFUA	Perfluoroundecanoic acid (PFUA) (ng/mL)	0.10
LBXNFOA	n-perfluorooctanoic acid (n-PFOA) (ng/mL)	0.10
LBXBFOA	Branch perfluorooctanoic acid isomers (Sb-PFOA) (ng/mL)	0.10
LBXNFOS	n-perfluorooctane sulfonic acid (n-PFOS) (ng/mL)	0.10
LBXMFOS	Perfluoromethylheptane sulfonic acid isomers (Sm-PFOS) (ng/mL)	0.10

References

- Benskin JP, De Silva AO, Martin JW. 2010. Isomer Profiling of Perfluorinated Substances as a Tool for Source Tracking: A Review of Early Findings and Future Applications. Rev Environ Contam Toxicol:111-160.
- Caudill SP, Schleicher RL, Pirkle JL. Multi-rule quality control for the age-related eye disease study. Statist Med 2008;27:4094-4106.
- Vyas SM, Kania-Korwel I, Lehmler HJ. 2007. Differences in the isomer composition of perfluorooctanesulfonyl (PFOS) derivatives. J Environ Sci Health A Tox Hazard Subst Environ Eng 42:249-255.
- Westgard JO, Barry PL, Hunt MR, Groth T. A multi-rule Shewhart chart for quality control in clinical chemistry. Clin Chem. 1981 Mar;27(3):493-501.

Codebook and Frequencies

SEQN - Respondent sequence number

Variable Name:	SEQN
SAS Label:	Respondent sequence number
English Text:	Respondent sequence number.
Target:	Both males and females 12 YEARS - 150 YEARS

WTSPF2YR - PFAS subsample weight

Variable Name: WTSPF2YR
SAS Label: PFAS subsample weight
English Text: PFAS subsample weight
Target: Both males and females 12 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
12575.1443 to 634502.90873	Range of Values	2551	2551	
0	No blood sample provided	1067	3618	
.	Missing	0	3618	

LBXPFDE - Perfluorodecanoic acid (ng/mL)**Variable Name:** LBXPFDE**SAS Label:** Perfluorodecanoic acid (ng/mL)**English Text:** Perfluorodecanoic acid (PFDeA) (ng/mL)**Target:** Both males and females 12 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
0.07 to 16.2	Range of Values	2551	2551	
.	Missing	1067	3618	

LBDPFDEL - Perfluorodecanoic acid Comment Code

Variable Name: LBDPFDEL
SAS Label: Perfluorodecanoic acid Comment Code
English Text: Perfluorodecanoic acid (PFDeA) Comment Code
Target: Both males and females 12 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
0	At or above the detection limit	1130	1130	
1	Below lower detection limit	1421	2551	
.	Missing	1067	3618	

LBXPFHS - Perfluorohexane sulfonic acid (ng/mL)**Variable Name:** LBXPFHS**SAS Label:** Perfluorohexane sulfonic acid (ng/mL)**English Text:** Perfluorohexane sulfonic acid (PFHxS) (ng/mL)**Target:** Both males and females 12 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
0.07 to 26.4	Range of Values	2551	2551	
.	Missing	1067	3618	

LBDPFHSL - Perfluorohexane sulfonic acid Comt Code

Variable Name: LBDPFHSL

SAS Label: Perfluorohexane sulfonic acid Comt Code

English Text: Perfluorohexane sulfonic acid (PFHxS) Comment Code

Target: Both males and females 12 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
0	At or above the detection limit	2500	2500	
1	Below lower detection limit	51	2551	
.	Missing	1067	3618	

LBXMPAH - 2-(N-methyl-PFOSA)acetic acid (ng/mL)**Variable Name:** LBXMPAH**SAS Label:** 2-(N-methyl-PFOSA)acetic acid (ng/mL)**English Text:** 2-(N-methylperfluorooctanesulfonamido)acetic acid (Me-PFOSA-AcOH)
(ng/mL)**Target:** Both males and females 12 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
0.07 to 3.8	Range of Values	2551	2551	
.	Missing	1067	3618	

LBDMPAHL - 2-(N-methyl-PFOSA) acetic acid Comt Code**Variable Name:** LBDMPAHL**SAS Label:** 2-(N-methyl-PFOSA) acetic acid Comt Code**English Text:** 2-(N-methylperfluorooctanesulfonamido)acetic acid (Me-PFOSA-AcOH)
Comment Code**Target:** Both males and females 12 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
0	At or above the detection limit	571	571	
1	Below lower detection limit	1980	2551	
.	Missing	1067	3618	

LBXPFNA - Perfluorononanoic acid (ng/mL)**Variable Name:** LBXPFNA**SAS Label:** Perfluorononanoic acid (ng/mL)**English Text:** Perfluorononanoic acid (PFNA) (ng/mL)**Target:** Both males and females 12 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
0.07 to 3.2	Range of Values	2551	2551	
.	Missing	1067	3618	

LBDPFNAL - Perfluorononanoic acid Comment Code

Variable Name: LBDPFNAL
SAS Label: Perfluorononanoic acid Comment Code
English Text: Perfluorononanoic acid (PFNA) Comment Code
Target: Both males and females 12 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
0	At or above the detection limit	2414	2414	
1	Below lower detection limit	137	2551	
.	Missing	1067	3618	

LBXPFUA - Perfluoroundecanoic acid (ng/mL)**Variable Name:** LBXPFUA**SAS Label:** Perfluoroundecanoic acid (ng/mL)**English Text:** Perfluoroundecanoic acid (PFUA) (ng/mL)**Target:** Both males and females 12 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
0.07 to 4.8	Range of Values	2551	2551	
.	Missing	1067	3618	

LBDPFUAL - Perfluoroundecanoic acid Comment Code

Variable Name: LBDPFUAL
SAS Label: Perfluoroundecanoic acid Comment Code
English Text: Perfluoroundecanoic acid (PFUA) Comment Code
Target: Both males and females 12 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
0	At or above the detection limit	656	656	
1	Below lower detection limit	1895	2551	
.	Missing	1067	3618	

LBXNFOA - n-perfluorooctanoic acid (ng/mL)

Variable Name: LBXNFOA
SAS Label: n-perfluorooctanoic acid (ng/mL)
English Text: n-perfluorooctanoic acid (n-PFOA) (ng/mL)
Target: Both males and females 12 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
0.07 to 19.5	Range of Values	2551	2551	
.	Missing	1067	3618	

LBDNFOAL - n-perfluorooctanoic acid Comment Code

Variable Name: LBDNFOAL

SAS Label: n-perfluorooctanoic acid Comment Code

English Text: n-perfluorooctanoic acid (n-PFOA) Comment Code

Target: Both males and females 12 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
0	At or above the detection limit	2532	2532	
1	Below lower detection limit	19	2551	
.	Missing	1067	3618	

LBXBFOA - Br. perfluorooctanoic acid iso (ng/mL)**Variable Name:** LBXBFOA**SAS Label:** Br. perfluorooctanoic acid iso (ng/mL)**English Text:** Branch perfluorooctanoic acid isomers (Sb-PFOA) (ng/mL)**Target:** Both males and females 12 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
0.07 to 4.8	Range of Values	2551	2551	
.	Missing	1067	3618	

LBDBFOAL - Br. perfluorooctanoic acid iso Comt Code

Variable Name: LBDBFOAL

SAS Label: Br. perfluorooctanoic acid iso Comt Code

English Text: Branch perfluorooctanoic acid isomers Comment Code

Target: Both males and females 12 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
0	At or above the detection limit	260	260	
1	Below lower detection limit	2291	2551	
.	Missing	1067	3618	

LBXNFOS - n-perfluorooctane sulfonic acid (ng/mL)**Variable Name:** LBXNFOS**SAS Label:** n-perfluorooctane sulfonic acid (ng/mL)**English Text:** n-perfluorooctane sulfonic acid (n-PFOS) (ng/mL)**Target:** Both males and females 12 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
0.07 to 144	Range of Values	2551	2551	
.	Missing	1067	3618	

LBDNFOSL - n-perfluorooctane sulfonic Comt Code

Variable Name: LBDNFOSL

SAS Label: n-perfluorooctane sulfonic Comt Code

English Text: n-perfluorooctane sulfonic acid (n-PFOS) Comment Code

Target: Both males and females 12 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
0	At or above the detection limit	2536	2536	
1	Below lower detection limit	15	2551	
.	Missing	1067	3618	

LBXMFOS - Sm-PFOS (ng/mL)**Variable Name:** LBXMFOS**SAS Label:** Sm-PFOS (ng/mL)**English Text:** Perfluoromethylheptane sulfonic acid isomers (Sm-PFOS) (ng/mL)**Target:** Both males and females 12 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
0.07 to 23.1	Range of Values	2551	2551	
.	Missing	1067	3618	

LBDMFOSL - Sm-PFOS Comment Code

Variable Name: LBDMFOSL

SAS Label: Sm-PFOS Comment Code

English Text: Perfluoromethylheptane sulfonic acid isomers (Sm-PFOS) Comment Code

Target: Both males and females 12 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
0	At or above the detection limit	2488	2488	
1	Below lower detection limit	63	2551	
.	Missing	1067	3618	